

Adroit Technologies

Protocol Driver

Name	OPC DA Client
DLL	OPC_CL



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1. Overview

OPC (OLE for Process Control) is a standard specification that makes it possible for automation/control software applications and field systems/devices to interoperate and communicate data between each other.

To learn more about what OPC is, visit the OPC foundation website <http://www.opcfoundation.org/>

The OPC Client driver for Adroit communicates to OPC servers that expose at least the OPC DA Custom Interface Version 1.0A. The driver is compatible with OPC DA Version 2.0. As of revision 56 of the driver, it is not version 3.0 compatible.



PLEASE NOTE

It is very important that Adroit users updating to revision 56 of the OPC client driver re-check all their OPC client device settings as things have changed since.

1. Read especially the setting "Override default DCOM security (for callbacks)" [below](#). By default, older versions of this driver had this setting hardcoded as enabled.
2. Also make a note of newly introduced [settings](#) (such as "auto recovery") and make sure you understand what is being provided for in this new version. "Auto recovery" was not a feature available in older versions of the driver.

1.1. Supported Hardware

This is equipment & vendor specific, installation instructions for specific OPC Servers is beyond the intended scope of this document, however [notes on specific differences](#) have been included. OPC Servers supporting version 2.0 of the OPC interface allow remote connections to servers as well as local OPC server connections.

Please see the section on "[Specific OPC Servers](#)", which covers differences experienced between commonly encountered OPC Servers.

Wiring

Not applicable.

2. Device Configuration

2.1. Driver Installation and Setup

The driver is installed in Adroit by default. If not, you will need to install it via the "Drivers" tab and click on "Add" in Adroit setup. The opc_cl.dll and the opc_cl.doc are then copied into the agent directory.

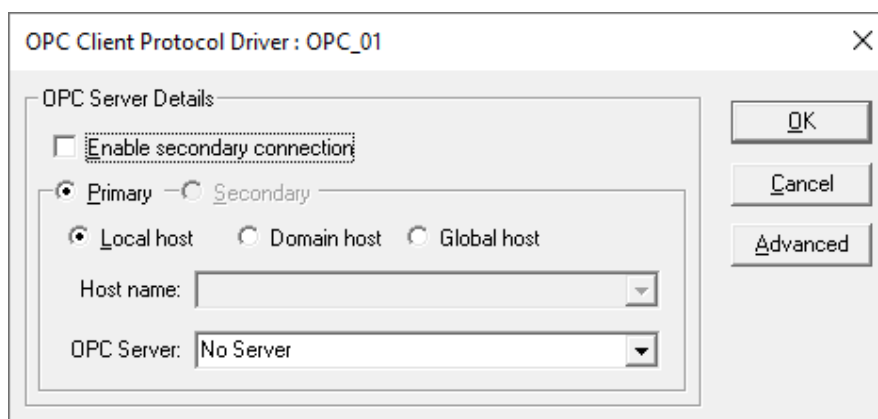
2.2. Device Configuration

2.2.1. Device Configuration

The configuration of an OPC client device has only three basic items of configuration for a connection. These are:

1. Remote or local
2. If remote, name of remote host (or PC)
3. The name of the OPC Server to connect to

As with most Adroit protocol drivers, you can also specify a secondary (or backup) connection.



Enable Secondary	Enables secondary configuration for dual redundant OPC Server Connections. If enabled, the primary and secondary buttons are also enabled to allow you to select the connection you wish to configure. This driver supports auto and manual resetting and switching of connections. See "Resetting and Switching Connections" below.
Local host	Connect to an OPC Server on the local PC. The "host name" dropdown box will be disabled.
Domain host / Global host	Select an OPC Server on a remote PC. The "host name" dropdown box will be enabled to allow you to select a remote PC. The reason for two buttons is simply to allow for fast and slow (respectively) network searching of remote PC's for populating the dropdown box. "Domain host" will search the local domain only while "global host" will take longer to search the wider network.
OPC Server	Select the OPC Server that you want to connect to. All servers registered on the PC will be available in the dropdown box. If you are not able to see OPC Servers on a remote PC please consult the paragraph on Security below.

2.3. Advanced Settings Dialog

All advanced settings are accessible in the advanced dialog; Changes to these settings are not required by default. Notable exceptions are listed in the section [OPC Server Notes](#).

OPC Client Protocol Driver : OPC_01 - Advanced settings (primary and secondary)

Mode settings

- ☐ Use synchronised polling
- ☒ Poll from device (as opposed to cache)
- ☐ Use OPC Server timestamps
- ☐ Do not validate items
- ☐ Use alternate item validation (add,remove)
- ☒ Force OPC Server to convert data types.
- ☐ Use OPC 1.0 interface

Maximum number of items per group/scan task

Boolean (1-3000)

Real/integer (1-1500)

Strings (1-100)

Read from device after write (0 to disable)

Read digital ms after write

Read integer ms after write

Read real ms after write

Read string ms after write

☒ Enable auto recovery

Interval: (10-3600 secs)

Item access path:

☒ Override DCOM default security (for callbacks)

HDA Connection (optional)

☒ Local host ☐ Domain host ☐ Global host

Host name:

HDA Server:

Use synchronised polling	(Default = off) OPC clients by nature receive data "by exception" from OPC Servers. However, you can use OPC for polling data from front end devices. This setting enables polling at the scan period specified when configuring scan items. This setting should only be used if necessary.
Poll from device	(Default = on) If polling is enabled, OPC can poll data from the device or internal cache of the OPC Server. Unless you are sure as to how your OPC Server operates, it is safer to leave this setting on.
Use OPC server timestamps	(Default = off) For OPC servers that provide timestamp data, whose timestamps must be reflected in Adroit as current and/or historical data, or else the current time is used for all data as it is received by the scanning subsystem. This setting is generally used for telemetry systems.
Do not validate items	(Default = off) This will skip the OPC item validation process. Normally used only if you configuring offline, appropriate if your OPC Server requires the actual hardware to be connected in order to validate correctly. Switching this setting off greatly speeds up the agent server loading process for configurations with large amounts of OPC scanning.
Use alternate item validation.	(Default = off) Some OPC servers have been found not to support the OPC "validate items" function. These servers can still be made to validate by turning this selection on which will cause the driver to add an item and remove and item for validation. Successfully adding the item implies a valid OPC item.
Force OPC to convert data types	(Default = on) The driver will ask the OPC Server for data in a format that best suits Adroit. If the option is off, the driver will accept any data types returned by the server and convert the data at runtime. Turn this option off if you have problems adding scan-points/items.

Use OPC 1.0	(Default = off) Forces the driver will use OPC interface version 1.0 if both are presented in the server. This setting allows operation where an OPC server has got version 2.0, but it does not operate satisfactorily.
Enable auto recovery	Enables the driver to attempt recovery of an OPC Server connection should the connection fail for any reason. This is achieved on a background thread for non-pollled devices OR on a read/write transaction for polled devices. If auto recovery is disabled, auto recovery will not take place regardless of whether a device is set for sync. polling or not, and regardless of whether you have a secondary connection configured or not. Therefore, if you have a secondary connection, ensure that auto recovery is enabled to ensure switching between connections. The auto recovery mechanism will always attempt to reconnect to the primary connection before trying the secondary connection.
Auto recovery interval	(Default = 30 seconds) The interval time at which attempts to recover a failed OPC server connection will be made.
Access path	(Default = none) The access path is an optional piece of information that can be provided by the user if supported by the OPC Server. For those of you familiar with the OPC specification, this is the same as the OPC item access path as discussed in the specification. If used, only items that use this access path may be scanned to this device. Its use is highly server specific but it is intended to allow the client to provide a 'recommendation' to the server regarding how to get to the data.
Maximum number of items per group/scan task	(Defaults - 500 for Booleans, 250 for Analogs and 100 for Strings) OPC uses groups to contain scan items. The number of items per group is controlled with these settings. A new group will be created once the max items for that particular type has been exceeded.
Read from device after write (0 to disable)	(Default = all zero) By default, Adroit subscribes to items in the OPC Server (i.e., is informed when data changes), the driver does not poll for data. To ensure that a successful write has taken place, the user can enable a "read after write" transaction by specifying a non-zero value in one or more of these fields. The driver will wait for the configured period and then executes an OPC read for that tag.
Override default DCOM security (for callbacks)	Overrides DCOM configuration settings for OPC Server callbacks to OPC client devices. This is a global setting for all OPC client devices. This setting is not enabled by default to allow your PC's DCOM configuration to be in control. Typically, if you can connect to an OPC Server, browse OPC items and select them for scanning, but no values are returned when scanning is started, reverse this setting to see if it makes a difference. If reversing the setting fixes the problem, make sure all other OPC devices are still working normally. If not, you need to leave this option unchecked and resolve the problem via the default DCOM configuration settings.

Registry Details

Not applicable.

3. Protocol Definition and Considerations

3.1. Scanning Configuration

The default address is *First:OPC:Address:01* this field must be substituted by the item ID (or name) of the OPC item being scanned. Item ID can be browsed using the browse button which launches Adroit OPCBROWSER utility to display all OPC items available on the selected OPC Server.

Note that some OPC Servers do not allow address space browsing, in such cases you need to type in a tag name manually. You then need to read the documentation of that server to get details of the item ID's, and how to manually type them in. It is always possible to manually enter an item, if the item is incorrectly typed, it may or may not be rejected based on the Server and the [Advanced settings](#) for Validation of item-IDs

The update rate of a tag is determined by the scan period specified. The driver creates new OPC groups automatically; a new group is created and managed for you if:

1. The maximum number of tags per group (as configured in the [Advanced dialog](#)) is reached.
2. Tags have a different scan rate.

3.2. Resetting and Switching Connections

This driver supports the resetting and switching of connections using the device agent's "switchconnection" and "resetconnection" slots. Setting either of these slots to 1 will cause the device to reset (close and re-open) the connection to the OPC Server or switch the connection from the primary to the secondary connection.

These slots must be used with care in a cluster as they are shadowed to the partner. To prevent shadowing, use the buttons as provided in the advanced dialog accessed from the configurator's scanning dialog – this will then perform the action on the connected agent server (master or standby) only.



WARNING

The OPC client driver uses "connection pooling". This means that OPC client devices that use the identical primary and secondary connection paths (host and OPC Server) will interact with the OPC server via a single OPC client connection. Therefore, when resetting or switching a device, any other devices using the same "pooled connection" will also be affected.

3.3. Message Protocol

Datascopes messages are displayed in ASCII mode. (Press the "A" key)

The underlying communication protocols to the device is not available to the OPC Client and thus not showed, see the [Troubleshooting](#) section.

This window will however show all validation of tags. If the tag cannot be validated, a message indicating the error code & message will be displayed here.

All item updates (process-data changes) are also indicated here with an offset number of the tag that's being updated.

3.4. Security

More often than not, OPC client applications fail to connect to remote OPC servers for all sorts of DCOM and NT Security reasons.

Firstly, it is recommended that OPC clients and OPC server PCs are configured within the same domain to ease security issues. If this is not the case, then you are required to ensure that users (and their passwords) are configured on all these PC's and kept synchronised.

3.4.1. DCOM Configuration

To set up the minimum-security run DCOMCNFG.EXE and execute the following (methodology for NT4.0 and Win2K only, as WinXP has a different interface described latter under "Windows XP Interface"):

1. On OPC Client machines, set the Default Authentication Level to (None) on the Default Properties tab.
2. On OPC Server machine, set the Default Authentication Level to (None) on the Default Properties tab, add "Everyone" to both the Default Access Permissions and Default Launch Permissions Access Control Lists on the Default Security tab.

With these settings, the domain setup (if any) as well as the IDs of the login users of each machine will be of no importance.



PLEASE NOTE

In very rare cases, some 3rd party OPC Servers/Clients may enforce security (by calling CoInitializeSecurity) explicitly - this will override the above settings which will not allow them to work. This is not a problem with Adroit OPC components.

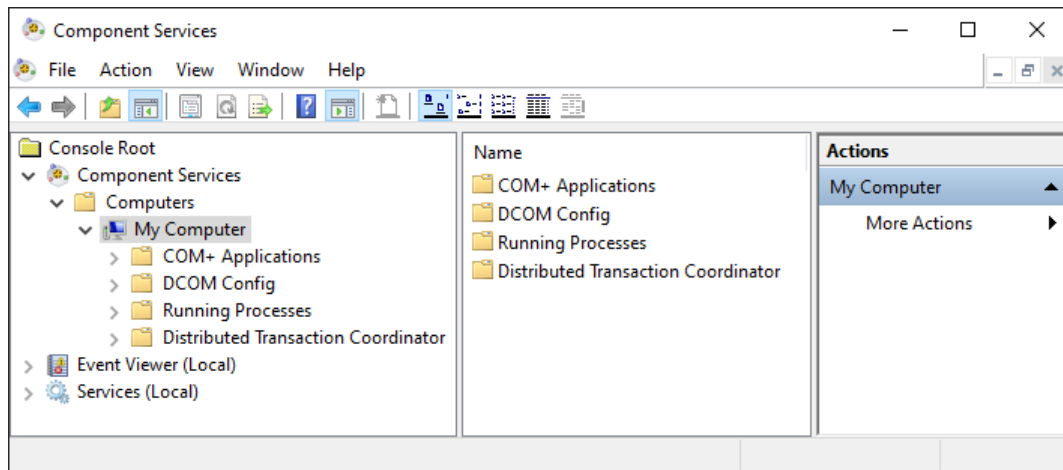
Finally, you may find that an OPC client application such as Adroit Setup program (PSEXEC.EXE) may be unable to browse OPC servers on some remote PC's. If this is the case, then ask your network administrator to do the following on the remote PC in question:

1. Run DCOMCNFG.EXE
2. Select the OPCENUM application on the "Applications" tab
3. Press the "Properties" button
4. Select the "Identity" tab which allows you to select the user account to be used when the OPCENUM application is run
5. Select "This user" and enter an administrator's username and password. Domain usernames must be entered as "domainname\username".
6. Close DCOMCNFG.EXE

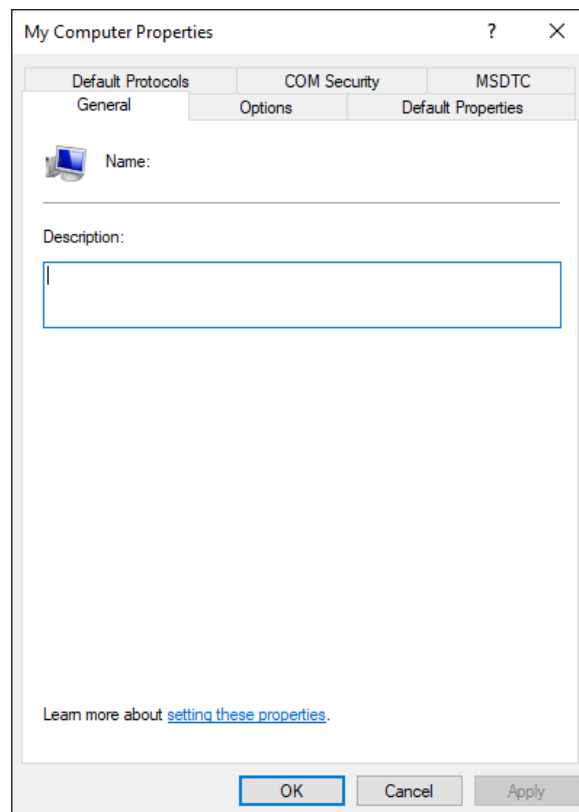
This should resolve the OPC Server browsing problem on a PC from remote OPC client applications.

3.4.2. Windows 10 Interface

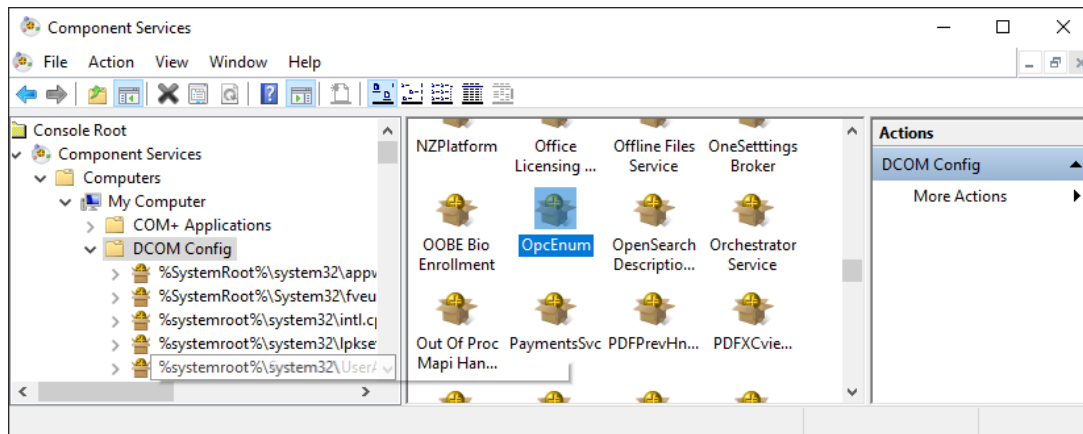
For Windows XP onwards, the DCOM config program is an MMC snap-in. Run it in the same way and get the screen below.



Navigate to "My Computer" as shown, and then right-click for properties on the "My Computer" branch.



From here the "Default Properties". "Default Protocols" and "COM Security" tabs present all the options that need changing. To edit the properties for a specific server as in the past, select the Program or server from the right-hand panel as shown below for OPCEnum.



4. General Notes and Troubleshooting

4.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

4.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

4.3. Troubleshooting

The Adroit Protocol Driver Monitor will give feedback on the OPC activity with error codes. OPC Servers supporting only version 1.0 of the OPC spec/interface do not allow remote DCOM clients i.e., OPC Server must be on same machine as Adroit.

If you have trouble getting a remote connection to work:

1. Use Explorer to connect to a share on the server machine. (Check your network)
2. Run Adroit's OPC browser and see if it connects to the remote machine and shows the OPC servers installed on it. (See the security section)
3. Try the same from a different computer, or with a local connection first by installing Adroit on the server machine.

What version of OPC Client driver do I have?

If using versions prior to 5.0.4.0:

1. Using Explorer, open your Adroit folder, and find opc_cl.dll
2. Right-click the file, select properties from the pop-up menu.
3. Click on the "Version" (it's the second one) tab and select the field "DriverRevision" from the list on the left. The driver's revision number will appear next to it.

In Adroit 5.0.4.0 onwards:

1. In the setup program drivers tab, select the Driver (not a device) from the tree.
2. One of the fields in the list to the right will be Revision.
3. The Adroit website always shows the latest driver revision currently in production.

Changing the settings in the "Advanced" settings for the device

This can resolve many configuration problems, please read the [Device configuration](#) and [OPC Server notes](#) sections.

Remember that any setting change requires a re-start of the Agent Server. Certain OPC Servers implement the OPC spec differently, Adroit Technologies development department attempt to

accommodate unexpected data format and custom-interface changes etc. by releasing driver updates. Check our website for driver patches. Also check out the next section [OPC Server notes](#) for your server.

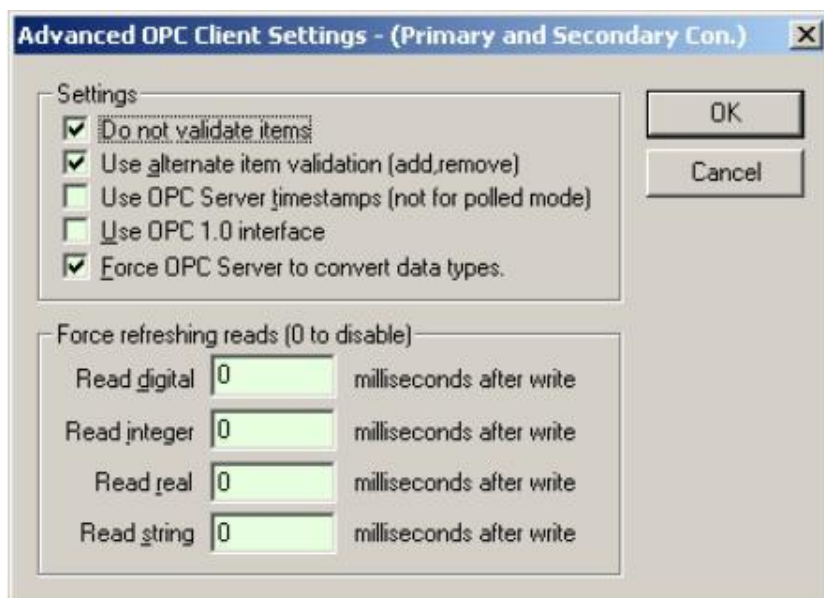
5. Appendix A - OPC Server Notes

5.1. Specific OPC Servers

OPC servers all behave differently; this section contains hints, notes, and generic problems sometimes experienced. Sometimes these issues are resolved in future revisions of these OPC Servers, and scenarios shown here may apply to servers we have not encountered or mentioned.

RS-LinX OPC Server	Rockwell	All Allen Bradley equipment and networks
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1. In the OPC server, you must create a topic for each CPU or unit that Adroit must communicate with. Simply put:
 - Open the RSLinx Window, select the CPU you want from the tree
 - Select "Add DDE/OPC topic" from the menu and provide a logical name, this logical name will appear in the OPC browser. You may use the same topic name as your adroit device name here.
2. CPUs without topics cannot be accessed via OPC.
3. **When configuring the OPC driver, you must select the OPC server, no other settings are needed initially. Be careful to select RSLinx OPC server, not "RSLinx Remote OPC server. The "Remote" is only used if the Agent server is on a different PC to the actual communications server**
4. Do NOT type in an access path, ensure this driver setting is empty.
5. **When connecting to SLC 500, 5/01 and 5/02 I/O are not numbered in files as laid out in program logic, but always start at 0.**
6. After all scan-points have been added, turn on the tick box for "do not validate items" in the advanced settings, and restart the machine. If Adroit starts up at a time when the PLC is disconnected, this prevents the loss of scan-points to disconnected devices which will occur.



When browsing tags in the OPC browser, ensure that the tick-box for "Detailed" mode is on.



INAT OPC	INAT	H1, and Profibus networks for Siemens S5 and S7 equipment
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1. Turn on the "do not validate items" tick-mark after commissioning all scans, since this server is sometimes slower to start, and may cause scan-points to be lost during starting on large systems.
2. In order to give the OPC server enough time to load, use **DeviceStartDelay** values of 5000ms or greater, to find out how to edit this registry setting, search for DeviceStartDelay in the Adroit help.

Johnson Controls OPC Server	Johnson Controls	Building management
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This OPC Server makes use of the @ symbol in the item-address, and since Adroit previously reserved the use of this symbol for subscription-scanning, you will need to disable subscription-scanning on this device. Adroit will then treat the @ symbols as part of the OPC item-address, and not as an internal agent tag-name reference.

Adding the REG_DWORD registry value "DisableAtScanning" to the registry for the required device and setting it to 1.

For example, if you wanted to disable @ scanning in the WESTDAM device-agent

1. Open the registry editor
2. Find the following registry key: HKEY_LOCAL_MACHINE\SOFTWARE\Adroit Technologies\Adroit\ProtocolDrivers\OPC_CL\WESTDAM
3. Add a DWORD entry called 'DisableAtScanning' (no quotes)
4. Set the value to 1
5. Perform steps 2 through 4 for any other Johnson Controls server devices.

Applicom OPC Server	Woodhead Systems Intl.	Communications server supporting many devices
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If you want to browse variables by name in a **Rockwell LogiX 5000** project, you will need to import the project into Applicom OPC, please read your Applicom OPC Server manual on how to do this.

6. Appendix B - DCOM Hardening

6.1. Background

The following section is extracted from the Adroit Knowledge Base regarding the security changes with regards to DCOM, referred to in the Microsoft community as DCOM Hardening.

<https://adroittechnologiesautomation.com/t/opc-dcom-hardening-microsoft-security-update/1271>

6.2. DCOM Hardening

To address the security vulnerability CVE-2021-26414 as found at <https://msrc.microsoft.com/update-guide/vulnerability/CVE-2021-26414> 6, Microsoft is hardening the Distributed Component Model (DCOM) on its Windows operating systems. DCOM Hardening means that an authentication of level RPC_C_AUTHN_LEVEL_PKT_INTEGRITY or higher for access to OPC Servers will be enforced. This level authenticates credentials and verifies that no call data has been modified in transit. See <https://docs.microsoft.com/en-us/windows/win32/rpc/authentication-level-constants> 3 for more information about authentication levels.

Currently, hardening is disabled by default but can be enabled via a DWORD registry value HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Ole\AppCompat\RequireIntegrityActivationAuthenticationLevel.

- If this registry value does not exist DCOM Hardening is disabled by default
- If this registry value equals 0, DCOM Hardening is disabled
- If this registry value equals 1, DCOM Hardening is enabled

For those with updated versions of one of the Windows Operating Systems as shown at <https://support.microsoft.com/en-us/topic/kb5004442-manage-changes-for-windows-dcom-server-security-feature-bypass-cve-2021-26414-f1400b52-c141-43d2-941e-37ed901c769c> 15, DCOM hardening will automatically apply on June 14 2022 where hardening changes will be enabled by default. This means that Microsoft will automatically generate this key on June 14 2022 and set it's value to 1. The effect of this is that users will begin to see unsecure connections to OPC Servers failing. This will ONLY happen if customers have not configured for secure connections to their OPC Servers. These customers will have two options

1. Set the above DWORD registry value HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Ole\AppCompat\RequireIntegrityActivationAuthenticationLevel to 0 and reboot your PC. This will only be effective until 14 Mar 2023
2. Configure for secure connections to OPC Servers using DCOM Configuration (Start/Run/dcomcnfg)
Adroit strongly recommends that all customers review their DCOM Configuration settings for all their OPC Servers and set the authentication level to "Packet Integrity" as shown below for the Adroit OPC Server.

Adroit.OPCServer Properties ? X

General Location Security Endpoints Identity

General properties of this DCOM application

Application Name: Adroit.OPCServer

Application ID: {1a863221-99df-4bb3-ba9a-96b78016c610}

Application Type: Local Server

Authentication Level: Packet Integrity ▼

Local Path: W:\adroit.deb64\adropcsvr.exe

Learn more about [setting these properties](#).

OK Cancel Apply

7. Appendix C – Terminology

COM	Component Object Model, basically a technique that allows developers to use a fixed specification to create inter-operable software.
DCOM	Distributed COM, allowing COM to work remotely between PCs. (see COM.)
Group	A collection of OPC items forms a group. Our OPC client-driver forms groups based on the Adroit tag-types (Boolean, Integer, Real and String) and on the update rates or scan-intervals. A group is analogous to a scan-task.
Item	A single OPC element, it has quality, a timestamp and a value, which are all reflected in the Adroit tag scanned to the item.
Item ID	This is the "address" that uniquely identifies an OPC Item or element. An Item ID typically uses a notation that describes the item based on it's parent using a dotted notation.
OLE Type	The OLE Type or canonical type is the presentation or original data format, which can be a bit (VT_BOOL), a 16-bit integer (VT_I2) amongst others, that the data was in. The client driver will convert types automatically, although this can be disabled to prevent conversion problems that occur when for instance converting a integer value into a bit. (See the Advanced setup option: option "Force OPC Server to convert types"). The type of an item can be seen in our OPC Browser program by "test"-ing the item.
OPC Foundation	An organization dedicated to distributing information about OPC itself, certification of OPC Servers and finding OPC Servers. http://www.opcfoundation.org
HDA	Historical Data Access – This specification, part of version 3.0 of OPC is not currently supported.
OPCAE	OPC - Alarming and Eventing interface – this specification, part of version 3.0 of OPC is not currently supported.

8. Revisions

8.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1	09-Jan-2022	K. Robinson	Changes to format, updated image resources. Added Appendix B DCOM Hardening. Removed Windows XP section	D. Jordaan
2.0				
3.0				
4.0				
5.0				
6.0				

8.2. DLL Revision

Rev.	Date	Changes
	4-Jun-02	Overview, screenshots updated. Troubleshooter updated.
	28-Aug-04	Server-specifics section added.
	29-Sept-04	Windows XP SP2 sub-section added to the Security section, Windows-XP MMC snap-in for DCOMCNFG described. A TOC was added at the top to simplify navigation. Terminology section added.
	21 Sept 2005	Revisions for driver file revision 56: Windows XP SP2 sub-section added to the Security section, Windows-XP MMC snap-in for DCOMCNFG described. A TOC was added at the top to simplify navigation. Terminology section added.
	5 April 2006	Johnson Controls OPC server + Applicom OPC server notes added. Contents Table rebuilt. Terminology section updated; more hyperlinks added throughout the document.
88	2009-2023	Various enhancements and improvements to the driver.